**25900285** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) Regular EXAMINATION October 2025

## Third SEMESTER

## Discipline Specific Core Course (DSC) - MG3DSCCHE202 - INORGANIC AND ORGANIC CHEMISTRY

(2024 ADMISSION ONWARDS)

**Duration: 1 Hour 30 Minutes** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. In vitamin $B_{12}$, cobalt is coordinated in a: [K] / [CO2]

   a). Heme group b). Corrin ring

   c). Porphyrin ring d). Chlorin ring

2. A food product label shows that it contains BHA. What purpose does it serve in the food? [U] / [CO4]

   a).Enhances color b).Improves flavor c).Prevents oxidation of fats d).Increases sweetness

3. Why is pyrrole considered an aromatic compound despite having a lone pair of electrons on nitrogen and being a five-membered ring? [E] / [CO3]

   a). The lone pair on nitrogen participates in the aromatic sextet b). It has alternating double and single bonds

   c). It is planar structure d). All of the above

4. Which molecule provides energy for the sodium-potassium pump? [K] / [CO2]

   a).ATP b).NADH

   c).Glucose d).$FADH_2$

5. The process of fruit ripening is chiefly regulated by the gas [K] / [CO4]

a). Carbide b). Ethylene

c). Methane d). Nitrogen

6. How many Optically active isomers are there for Lactic acid? [U] / [CO5]

a). 1 b). 2

c). 3 d). 4

7. What is the main principle used in a nuclear power reactor? [U] / [CO1]

a). Fission b). Fusion

c). Radioactivity d). None of them

8. Stereoisomers of a compound which are not mirror images of each other are known as [K] / [CO5]

a). Enantiomers b). Conformers

c). Diastereomers d). Configurations

9. A compound which has the ability to sustain an induced ring current is called [K] / [CO3]

a).diatropic b).isotropic

c).anisotropic d).protropic

10. [K] / [CO4]

Fast foods are often high in:

a).Vitamins and minerals b).Fibre and antioxidants c).Saturated fats, sugar, and salt d).Complex carbohydrates

**[1x10 = 10]**

## Part B

Short Answer Questions

Answer any **4** questions

Each question carries **3** marks

11. How can you detect the presence tamarind or date seed powder in coffee powder? [U] / [CO4]

12. Summarise the role played by haemoglobin in oxygen transport [K] / [CO2]

13. Explain geometrical isomerism of aldoximes and ketoximes [U] / [CO5]

14. Explain critical, subcritical and supercritical mass [U] / [CO1]

15. Draw the Sawhorse and Newmann projection of eclipsed conformation of ethane [K] / [CO5]

16. Draw different isomeric structures of Xylene [U] / [CO3]

**[3x4 = 12]**

## Part C

Short Essay Questions

Answer any **2** questions

Each question carries **7** marks

17. What is artificial sweeteners with examples and write the qualities of an ideal sweetener? [K] / [CO4]

18. The cis isomer normally has a higher dipole moment than the corresponding trans isomer . Justify with suitable examples. [A] / [CO5]

19. Differentiate the structural differences between hemoglobin and myoglobin. [U] / [CO2]

**[7x2 = 14]**

## Part D

Essay Type Questions

Answer any **1** questions

Each question carries **14** marks

20. Compare the basicity of pyrrole, pyridine, and indole. Explain with molecular orbital considerations and examples. [U] / [CO3]

21. Elaborate on the use of radioisotopes in various fields. [K] / [CO1]

**[14x1 = 14]**